

Project: Arduino-based Greenhouse Manager

1. Introduction

The purpose of this document is to outline the specifications, constraints, and requirements for the development of an Arduino-based Greenhouse Manager. The Greenhouse Manager is intended to automate and optimise the control and monitoring of environmental conditions within a greenhouse to enhance plant growth.

2. Objectives

The primary objectives of the Greenhouse Manager project are as follows:

- **Automated Environmental Control:** Implement an Arduino-based system to automate and control environmental parameters such as temperature and humidity within the greenhouse.
- **Sensor Integration:** Integrate sensors for real-time monitoring of key environmental factors to ensure optimal conditions for plant growth.
- **Modular Design:** System design should allow for the user to use their personal choice of misters, fans, valves, etc.
- **Watering System:** Implement an automated watering system based on timing; allowing precise control over plant hydration.
- **User Interface:** Develop a user-friendly interface for the greenhouse manager to allow manual control and monitoring, as well as access to historical data.

3. Constraints

The following constraints must be considered during the development of the Greenhouse Manager:

- **Material:** The project must include an arduino board.

- **Time:** First prototype should be done within one week on 2/9/24.

4. Requirements

4.1 Functional Requirements

1. **Temperature Control:**

- The system should maintain the greenhouse temperature within a specified range.
- Users must be able to set the desired temperature through the user interface.

2. **Humidity Control:**

- The system should regulate humidity levels within predefined limits.
- Users must have the ability to adjust humidity settings.

2. **Water Control:**

- Must be able to talk with valves to water plants.
- Allow users to set watering schedules.

3. **Sensor Integration:**

- Integrate temperature, humidity, and watering sensors for real-time monitoring.
- Ensure accurate and reliable sensor readings.

4. **User Interface:**

- Develop a user-friendly interface for manual control and monitoring.
- Include visual representations of current environmental conditions.

4.2 Non-Functional Requirements

1. **Reliability:**

- The system should operate reliably in a greenhouse environment for extended periods.

2. **Scalability:**

- Design the system architecture to accommodate different greenhouse sizes.

3. **Ease of Maintenance:**

- Ensure easy maintenance with accessible components and clear documentation.

5. System Architecture

